

Ex. no. 5 Study of Activation Function and their Role

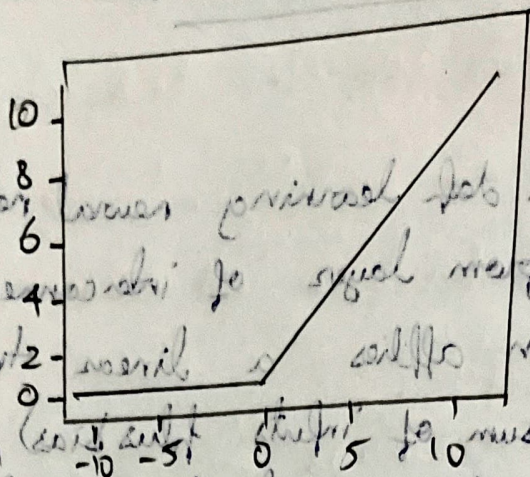
Introduction

In deep learning neural networks are built from layers of interconnected neurons. Each neuron applies a linear transformation (weighted sum of inputs plus bias), followed by a non-linear activation function. Without activation function, neural network function would be equivalent to a linear model, regardless of depth.

Need for Activation function:

- * Introduce Non-linearity: Real world data are highly non-linear. Without activation, the neuron cannot capture such complexity.
- * Enable Deep Architecture: Multiple layers with activation allow networks to learn hierarchical representation.
- * Control Output Range: They shape neuron output (eg, bounded between 0 and 1, -1 and 1, or unbounded).
- * Gradient Flow: They affect how errors backpropagate during training.

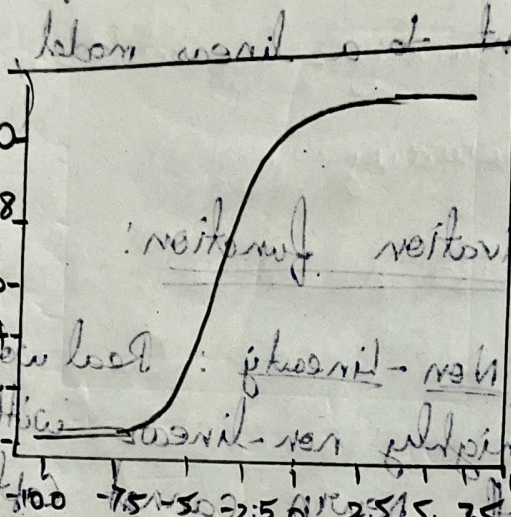
Role of Activation Function and its Role



ReLU

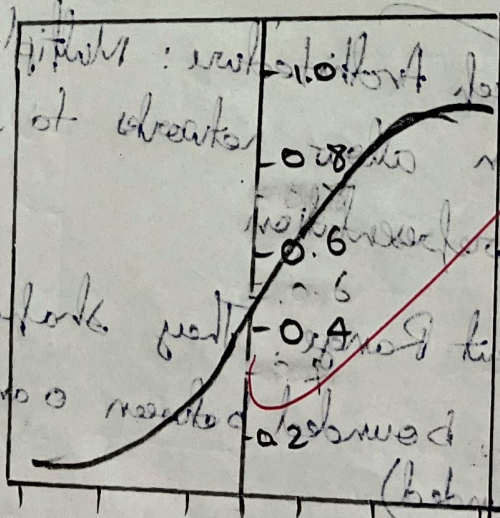
In self-driving cars, ReLU is used to process sensor data. For example, it can be used to detect if a pixel in an image is above a certain threshold, indicating a potential obstacle.

Rectified Linear Unit



Sigmoid

The Sigmoid function is often used in the output layer of a neural network to produce a probability value between 0 and 1. It is also used in hidden layers for binary classification tasks.



Softmax

The Softmax function is used in the output layer of a neural network to produce a probability distribution over multiple classes. It ensures that the sum of the probabilities for all classes is equal to 1.

Common Activation Functions

(a) Rectified Linear Unit (ReLU)

$$f(x) = \max(0, x)$$

- if the input is positive it passes it through
- if the input is zero or negative, it turns it into 0

(b) Sigmoid

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

- For very large negative $x \rightarrow e^{-x}$ is huge then output ≈ 0

- For $x=0 \rightarrow \text{output} = 0.5$

- For very large positive $x \rightarrow e^{-x}$ is tiny $\rightarrow \text{output} \approx 1$

Range = $[0, 1]$

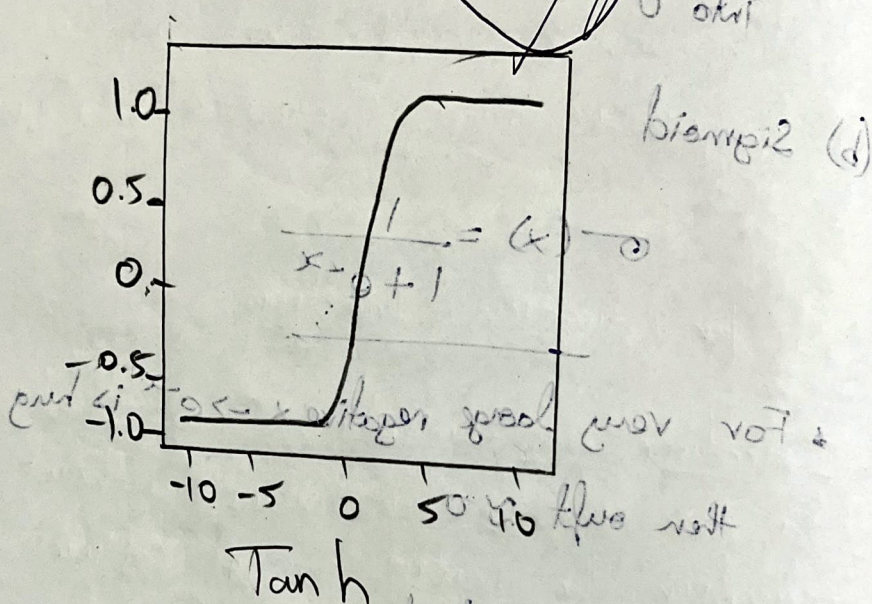
(c) Softmax Function

$$\sigma(z)_i = \frac{e^{z_i}}{\sum_j e^{z_j}}$$

- The outputs are probabilities (sum=1)
- The largest input gets the highest probability, but others aren't discarded completely

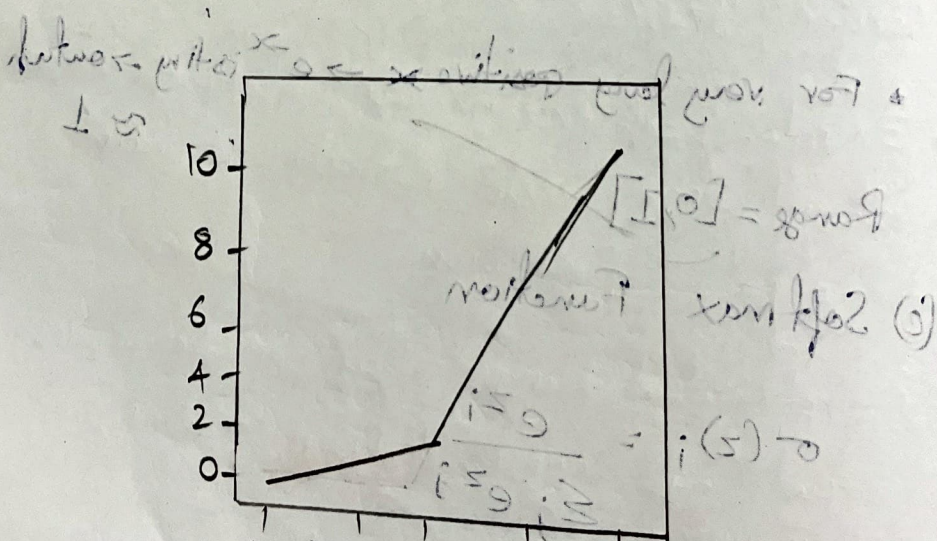
(a) Rectified Linear Unit (ReLU)
 common Activation functions
 $f(x) = \max(0, x)$

if the input is positive it passes it through
 + if the input is zero or negative it turns it
 into 0



Tanh

For $x=0 \rightarrow \text{output} = 0.2$



The output is always $\in [0, 1]$

ReLU
 The output is always $\in [0, \infty)$
 The output is always $\in [0, \infty)$
 The output is always $\in [0, \infty)$

Role of Activation Function

1. Add non-linearity
2. Control output range
3. Aid gradient flow
4. Make layers useful
5. Learn complex patterns.

d.) Hyperbolic tangent (tanh)

$$f(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

Range: $(-1, 1)$

↳ Centred around 0, better than sigmoid.
Still suffers from vanishing gradient.

e.) Leaky ReLU

$$f(x) = x \text{ if } x > 0 \text{ else } 0.01x$$

↳ Prevents dead neurons by allowing a small gradient for negative inputs.

Pseudo Code

- ①. Initialize dataset
- ②. Define a simple neural network with:
 - Input layer
 - output layer
 - hidden layer

3. choose an activation function for the hidden layers (sigmoid, tanh, ReLU, Leaky ReLU, Softmax)

4. For each activation function:

a. Assign the activation function to the network

b. Initialize weights and biases

c. Define loss function

d. Define optimizer

e. for fixed number of epochs:

i. Forward pass

ii. Compute loss

iii. Backward pass

iv. update weights

f. evaluate the trained model on test data

Observation:-

↳ Activation functions introduce non-linearity

↳ They control the flow of information through the network by transforming weighted inputs

↳ The choice of Activation function affects training speed, gradient flow, and final accuracy

Result:-

Activation functions are essential in deep learning as they introduce non-linearity enabling neural network to learn complex patterns.